

1. Determine the following

a. $17 \bmod 2$

(answers may vary) $17 \bmod 2 = 1$. More generally, $17 \bmod 2 = 2k + 1$ for any $k \in \{0, 1, 2, 3, \dots\}$ since $2 \mid 17 - (2k + 1)$.

b. $\text{lcm}(7^{4000} \cdot 5^2 \cdot 2^4, 7^4 \cdot 5^{1000} \cdot 2)$

$$\text{lcm}(7^{4000} \cdot 5^2 \cdot 2^4, 7^4 \cdot 5^{1000} \cdot 2) = 7^{4000} \cdot 5^{1000} \cdot 2^4$$

c. $17 = x \cdot 2 - y \cdot 15$. What is $\gcd(x, y)$?

The $\gcd(x, y)$ is the least positive integer representable in the form $x \cdot s + y \cdot t$. All others are *multiples* of $\gcd(x, y)$, e.g., $2 \cdot 1 + 6 \cdot 1 = 8$, $\gcd(2, 6) = 2$ and $2 \mid 8$.

Since x and y can be written as a linear combination summing to 17, the $\gcd(x, y) =$ either **1** or **17**.

2. Complete the following

a. Write the given set in tabular form: $A = \{x \mid x \in \mathbb{N} \text{ and } x \leq 5\}$

$$A = \{1, 2, 3, 4, 5\}$$

b. Write the given set in set-builder form: $A = \{\dots - 2, 0, 2, 4\}$

$$A = \{x \mid x \text{ is an even integer and } x \leq 4\} \text{ or } A = \{x \mid x = 2n \leq 4 \text{ and } n \in \mathbb{Z}\}$$

3. Let $A = \{1, 2, 3\}$, $B = \{2, 4, 6\}$, $C = \{4, 5, 6\}$.a. Find $A \cup C$

$$A \cup C = \{1, 2, 3, 4, 5, 6\}$$

b. Find $A \cap B$

$$A \cap B = \{2\}$$

c. Find $A \cap C$

$$A \cap C = \emptyset$$

d. Find $(A \cup C) \cap B$

$$(A \cup C) \cap B = \{1, 2, 3, 4, 5, 6\} \cap B = \{2, 4, 6\}$$

4. Let A and B be two sets. Prove that $A \cap B \subseteq A$.

Proof. Let $x \in A \cap B$. Then $x \in B$ and $x \in A$. Thus $x \in A$ and $A \cap B \subseteq A$.

□

5. Let $A \neq \emptyset$ be a subset of B . Prove that $A \cup B \neq A$.

Proof. Let $A = \{a\}$ and $B = \{a, b\}$. Then $A \subseteq B$, $A \neq \emptyset$, but $\{a, b\} = A \cup B \neq A = \{a\}$.

□

6. Give the definition of the following:

a. A *function*

A *function* is a relation that pairs element from a set A to exactly one element in a set B .

b. An *onto function*.

A function $f : A \rightarrow B$ is *onto* if for every $b \in B$ there exists an $a \in A$ such that $f(a) = b$.

7. Consider the map $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x, y) = xy$. Then f is:

(a) an onto function.

(b) a 1-1 function.

(c) a bijection.

(d) none of the above.

f is onto since for every $b \in \mathbb{R}$, $f(b, 1) = b \cdot 1$. f is not 1-1 since $f(-2, -2) = 4$ and $f(2, 2) = 4$. So the answer is **(a)**.

8. Let $f : A \rightarrow B$ be a function, $A = \{1, 2, 3\}$, and $B = \{1, 2\}$.

a. Can f be 1-1? Explain.

No. $|A| > |B|$ so f cannot be 1-1 by the Pigeon Hole Theorem.

b. Show that f can be onto.

(answers can vary) Let $f(1) = 1$, $f(2) = 2$, and $f(3) = 1$.

Note that f does not have to be onto, e.g, let f be the constant function, $f(x) = 1$.

9. Define the following relation on the real number set: $x\mathcal{R}y \iff xy > 0$. Show the following:

a. $\mathcal{R}$ is *not* reflexive.

Proof. Let $x = 0$. Then $x \cdot x = 0 \not> 0$. So $\mathcal{R}$ is not reflexive. □

b. $\mathcal{R}$ is transitive.

Proof. Let $x, y, z \in \mathbb{R}$. If $x \cdot y > 0$ and $y \cdot z > 0$ then $xy/y = x > 0$ and $yz/y = z > 0$. Thus $x \cdot z > 0$. So $\mathcal{R}$ is transitive. □

c. $\mathcal{R}$ is *not* an equivalence relation.

Proof. $\mathcal{R}$ is not reflexive by part (1). So $\mathcal{R}$ is not an equivalence relation.

□

10. Give an example of a relation on a set which is an equivalence relation.

(answers may vary) Let $x\mathcal{R}y \iff x = y$.

$x = x$ so $\mathcal{R}$ is reflexive. $x = y \Rightarrow y = x$ so $\mathcal{R}$ is symmetric. If $x = y$ and $y = z$ then $x = y = z$ so $\mathcal{R}$ is transitive. Thus $\mathcal{R}$ is an equivalence relation.